

A Fibonacci age law for an injective relative of the $3x + 1$ map, and numerical evidence for an open question of Reyes Jiménez

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September 2026. Draft — not yet checked against all primary sources; see Section 6.

Abstract

Replacing the halving step $n \mapsto n/2$ of the $3x + 1$ map by $n \mapsto n - |n/2|$ leaves positive integers unchanged and turns the map on negative integers into the injective, magnitude-increasing map $m \mapsto 3m - 1$ (m odd), $m \mapsto 3m/2$ (m even). Its inverse is a terminating process, giving every integer an *age*. We prove that the ages are governed by Fibonacci numbers: $\Pr(\text{age} \geq t) = F_{t+3}/(2 \cdot 3^t)$, and more generally $F_{t+3}/(2q^t)$ when 3 is replaced by any odd $q \geq 3$. The Fibonacci structure comes from the prime 2 only. We show that a recent forward Fibonacci count of Reyes Jiménez is the 2-adic counterpart of this law and reprove it in a few lines, identifying the counted set as the integers whose initial Syracuse exponents are all at most 2. That paper asks whether every $n > 1$ eventually has two consecutive halvings. We observe that the $3x + 1$ conjecture implies this, that it fails for negative integers, and we verify it exhaustively for all $n < 2^{56}$ with an algorithm whose cost is $O(\varphi^B)$ rather than $O(2^B)$. Beyond an integer's own binary digits, the measured per-step survival probability is 0.809017, against $\varphi/2 = 0.809017$. The same method, run backwards in base 3, verifies a conjecture of Banerji for all odd $n < 2 \cdot 3^{32}$.

1 The map

Let $F(n) = 3n + 1$ for odd n and $F(n) = n - |n/2|$ for even n . For $n > 0$ this is the Collatz map. For $n < 0$ the even step is $n \mapsto 3n/2$, so on magnitudes $m = |n|$ the map is

$$G(m) = 3m - 1 \quad (m \text{ odd}), \quad G(m) = 3m/2 \quad (m \text{ even}). \quad (1)$$

For odd $q \geq 3$ and $c = \pm 1$ put $G_{q,c}(m) = qm + c$ for odd m and $qm/2$ for even m , so that $G = G_{3,-1}$. The orbit of 1 under G is 1, 2, 3, 8, 12, 18, 27, 80, 120, 180, 270, 405, 1214, ...

The closest named relative is the map $x \mapsto (3x + 1)/2$ (x odd), $3x/2$ (x even) from Mahler's Z-number problem [10]. The map (1) differs in the odd branch, and that difference is what makes it injective.

2 Elementary properties

Proposition 1 *$G_{q,c}$ is injective and $G_{q,c}(m) > m$ for all $m \geq 1$. Its image is $\{m : q \mid m\} \cup \{m \text{ even} : m \equiv c \pmod{q}\}$, of density $3/(2q)$. For $q = 3$, $c = -1$ exactly half of the positive integers have no predecessor, namely those with $m \equiv 1 \pmod{3}$ or $m \equiv 5 \pmod{6}$.*

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Proof. $qm + c > m$ and $qm/2 > m$. Odd m map to even values congruent to c modulo q ; even m map to multiples of q . These two sets are disjoint, each branch is injective, and every element of either set is attained. $\square$

Proposition 2 (Valuation transfer) *Let b be odd and write $qb + c = 2^a u$ with u odd. The next odd value on the $G_{q,c}$ -orbit of b is $q^a u$, and $\gcd(u, q) = 1$. Hence its q -adic valuation equals $v_2(qb + c)$.*

Proof. Each of the a even steps replaces one factor 2 by one factor q . Since $qb + c \equiv c \pmod{q}$, u is prime to q . $\square$

Thus G retains, as a power of 3, exactly the power of 2 that the Syracuse map discards. In terms of prime factorisations, the even step of G exchanges a factor 2 for a factor 3 and preserves the number Ω of prime factors, whereas halving decreases Ω by one.

3 The age law

Backward steps from m are: **A**, $m \mapsto 2m/q$, available iff $q \mid m$; and **B**, $m \mapsto (m - c)/q$, available iff m is even and $m \equiv c \pmod{q}$. At most one is available. Let $\text{age}(m)$ be the number of backward steps that can be taken from m , and let the *age word* of m record them, most recent first. The last value reached, which has no predecessor, is the *seed* of m . For G : $\text{age}(27) = 6$, with word AAABAB and seed 1.

Write $F_1 = F_2 = 1$ for the Fibonacci numbers.

Theorem 3 *Let $q \geq 3$ be odd, $c = \pm 1$ and $t \geq 1$.*

1. *The age words are exactly the words over $\{\mathbf{A}, \mathbf{B}\}$ with no factor BB.*
2. *$\min(\text{age}(m), t)$ depends only on $m \bmod 2q^t$.*
3. *The number of residues modulo $2q^t$ with $\text{age} \geq t$ is F_{t+3} . Consequently the natural density of $\{m : \text{age}(m) \geq t\}$ is $F_{t+3}/(2q^t)$.*

Proof. (i) After a step B the value $(m - c)/q$ is odd, since $m - c$ is odd and q is odd; but B requires an even value. So B never follows B. After a step A the value $2m/q$ is even, so the parity requirement of B is automatic there. Hence the only parity condition in an entire backward walk is on m itself, and only if the first step is B.

(ii) Each backward step has the form $m \mapsto (um + v)/q$ with $u \in \{1, 2\}$, a unit modulo q . After i steps the ancestor is $(Um + V)/q^i$ with U a unit modulo q . The condition for step $i + 1$, that the ancestor be congruent to 0 or to c modulo q , therefore selects exactly one of the q lifts of $m \bmod q^i$ to $m \bmod q^{i+1}$. By induction every BB-free word of length t occurs, determines $m \bmod q^t$ uniquely, and distinct words determine distinct residues.

(iii) By the Chinese remainder theorem a word beginning with A corresponds to two residues modulo $2q^t$ (the parity of m is free), and a word beginning with B to one (m even). There are F_{t+1} BB-free words of length t beginning with A and F_t beginning with B, and $2F_{t+1} + F_t = F_{t+3}$. $\square$

Corollary 4 1. *The mean age is $\sum_{t \geq 1} F_{t+3}/(2q^t) = \frac{1}{2} \left(\frac{2+x}{1-x-x^2} - 2 \right)$ at $x = 1/q$; for $q = 3$ it equals $11/10$.*

2. The language of age words has factor complexity $p(n) = F_{n+2}$: it is the full golden-mean shift.
3. Every positive integer has unique coordinates (seed, age word), and the positive integers are a disjoint union of $G_{q,c}$ -orbits of seeds.
4. Forward, the run of even steps after an odd b has length $v_2(qb + c)$, which is geometric of ratio $1/2$ with respect to density. Backward, a run of steps A continues iff q divides the ancestor, which is geometric of ratio $1/q$.

The Fibonacci numbers here are produced by the prime 2 alone (an odd value cannot take the odd-type step); q enters only through the denominator.

Checks. Part (iii) was confirmed by exhaustive count for $q \in \{3, 5, 7, 9, 11\}$ and both signs of c , with $t \leq 9$ for $q = 3$; the mean age over $m \leq 2 \cdot 3^{11}$ is 1.09999; the run-length tails in Corollary 4(iv) agree to four decimals.

4 The forward counterpart

Let $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n . Reyes Jiménez [12, Theorem 4.7] proves that exactly F_{m+1} odd integers $n \in \{1, \dots, 2^m\}$ satisfy $T^i(n) \not\equiv 4 \pmod{6}$ for $1 \leq i \leq m - 1$, using the spectral radius φ of the modulo-6 transition graph with the vertex 4 removed.

Proposition 5 *For odd n , some $T^i(n)$ with $1 \leq i \leq m - 1$ is congruent to 4 modulo 6 if and only if the parity word of n of length m contains 00.*

Proof. After an odd step the value is $\equiv 2 \pmod{3}$. Halving such a value gives a value $\equiv 1 \pmod{3}$; values $\equiv 1 \pmod{3}$ arise in no other way, and no value after the first step is divisible by 3. A value is $\equiv 4 \pmod{6}$ iff it is $\equiv 1 \pmod{3}$ and even, that is, iff it was produced by a halving and the next step is a halving. $\square$

By the bijection of Terras [14] between parity words of length m and residues modulo 2^m , words that begin with 1 and avoid 00 correspond to F_{m+1} odd residues. This reproves [12, Theorem 4.7] and identifies the counted set: the n whose initial Syracuse exponents are all at most 2. We verified the equality of the two sets for $m \leq 14$.

	forward [12]	backward (Theorem 3)
forbidden factor	00	BB
digit bijection	parity word $\leftrightarrow n \bmod 2^m$	age word $\leftrightarrow m \bmod q^t$
count	F_{m+1} of 2^{m-1}	F_{t+3} of $2q^t$
status	selects a null set of orbits	holds for every integer

5 An open question of Reyes Jiménez

In [12, Section 5] it is asked whether every odd $n > 2$ eventually meets 4 $\pmod{6}$, with the remark that an explicit bound $m_0(n) \leq f(n)$ on the first such level would settle it. By Proposition 5 this is:

Question Q *Does every integer $n > 1$ have two consecutive halvings in its T -orbit? Equivalently, is some Syracuse iterate of every odd $n > 1$ congruent to 5 modulo 8?*

- Remark 6** 1. The $3x+1$ conjecture implies a positive answer. *The odd Syracuse predecessors of 1 other than 1 are $(4^j - 1)/3$ with $j \geq 2$, reached with exponent $2j \geq 4$.*
2. The statement fails for negative integers: -1 is fixed with exponent 1, and $-5 \mapsto -7 \mapsto -5$ uses exponents 1, 2. Any proof must therefore use positivity.
3. The 2-adic integers whose parity sequence avoids 00 form a set of Hausdorff dimension $\log_2 \varphi \approx 0.694$, because the parity-sequence map is a 2-adic isometry conjugating T to the shift [9].
4. An orbit giving a negative answer is either a non-trivial cycle all of whose exponents are at most 2, or is unbounded. Inside the family the exponent is 1 for $n \equiv 3, 7 \pmod{8}$ and 2 for $n \equiv 1 \pmod{8}$, so typical members grow; Question Q is of divergence type. The bounds on m -cycles of Simons-de Weger [13] and Hercher [6] do not apply, since such a cycle may have arbitrarily many local minima.

Algorithm

The residues r modulo 2^j whose first j steps avoid 00 form a tree with F_{j+1} nodes at level j . With each node carry $x = T^j(r)$ and 3^s , where s is the number of odd steps so far. Replacing r by $r + 2^j$ replaces x by $x + 3^s$, which has the opposite parity, so both children are obtained in $O(1)$: the odd child always survives, and the even child survives iff the previous step was not a halving. Walk the tree to depth B and finish each of the F_{B+1} leaves, which are genuine integers $r < 2^B$ with $T^B(r)$ already known, by direct iteration. Integers that leave the family earlier are pruned. The search is exhaustive for $n < 2^B$ at cost $O(\varphi^B)$.

Results

B	leaves F_{B+1}	time	longest survival	attained at n	steps/ $\log_2 n$
32	3 524 578	< 0.1 s	100	1 548 635 775	3.28
40	165 580 141	0.2 s	129	971 145 319 023	3.24
48	7 778 742 049	10.5 s	152	19 914 105 817 593	3.44
56	365 435 296 162	525 s	173	66 107 737 450 688 865	3.10

(Rust, 32 threads, 128-bit arithmetic with an overflow guard that never triggered; counts and records agree with an independent Python implementation for $B = 24, 32, 40$.)

Question Q has a positive answer for every n below the largest bound in the table. A model in which each further step is survived with probability $\varphi/2$ predicts record survival times $\log_2 n / \log_2(2/\varphi) \approx 3.27 \log_2 n$, and the observed records lie on that line. We conjecture $m_0(n) \leq 4 \log_2 n$ for all sufficiently large n .

Self-generated digits

For $B = 56$, among the $3.65 \cdot 10^{11}$ integers that survive their own 56 binary digits, the conditional probability of surviving one more step is 0.809017, 0.809017, 0.809012, 0.809021 at levels 56, 64, 72, 80; and $\varphi/2 = 0.809017$. Beyond level B the parity is no longer read from the digits of n but from digits produced by the iteration, and these leave the family at exactly the rate at which free digits would.

The backward counterpart

Banerji [3] conjectured that backward Syracuse iteration restricted to exponents at most 2 always reaches a multiple of 3. On odd n this iteration is the deterministic map $b(n) = (2n - 1)/3$ if $n \equiv 2 \pmod{3}$, $b(n) = (4n - 1)/3$ if $n \equiv 1 \pmod{3}$, with $n = 1$ fixed. The same algorithm applies with base-3 digits: replacing r by $r + 2 \cdot 3^j d$ changes $b^j(r)$ by a power of 2 times d , so exactly one of the three lifts dies, and the survivors of t steps are 2^t of the 3^t odd residues modulo $2 \cdot 3^t$. The conjecture holds for every odd $1 < n < 2 \cdot 3^{32} \approx 2^{51.7}$; the longest survival is 83 steps, at $n = 116\,393\,404\,689\,949$; the per-step survival probability is 0.66667 against $2/3$. The exceptional set is 3-adic, of dimension $\log_3 2$.

6 Related work, and what has been checked

Forward Fibonacci counts for the $3x + 1$ map are known: the set of itineraries of the unaccelerated map is the golden-mean shift [5]; Winkler [16] counts residue classes modulo powers of 2 by Fibonacci numbers; and [12] is discussed above. Albert, Gudmundsson and Ulfarsson [2] enumerate Collatz-induced permutations by Fibonacci numbers. That backward admissibility is decided 3-adically is the mechanism of Wirsching [15], for the map T , where runs of halvings are unrestricted and the counts are not Fibonacci. Step A of the inverse of G is the digit-0 edge of the base-3/2 number system of Akiyama, Frougny and Sakarovitch [1]; see also [4] for negative integers and 3-adic convergence. Merlini and Sala [11] obtain a Fibonacci-type recursion as a heuristic growth rate of the inverse tree.

We text-searched both annotated bibliographies of Lagarias [7, 8] and found no entry for the map (1), for an injective variant of this form, or for a Fibonacci count of backward residues. The sequence 1, 2, 3, 8, 12, 18, 27, 80, ... is not in the OEIS. **We have not read the primary texts of [15], [16], [11] or [3], only annotations and abstracts.**

Theorem 3 and Proposition 5 are elementary, and every ingredient is known. The prudent description is: an elementary backward, q -adic companion to the known forward Fibonacci counts, obtained by Wirsching's mechanism applied to an injective relative of Mahler's map.

7 Questions

1. Is there a single bijection explaining both Fibonacci counts, exchanging the roles of 2 and q ?
2. Does a single G -orbit have the Markov statistics ($\pi_A = 2/3$, $P(B \mid A) = 1/2$)? Three orbits of 200 000 steps pass a χ^2 test; a proof appears as hard as the distribution of $(3/2)^k$ modulo 1.
3. Can Question Q be settled for cycles alone?

Acknowledgements and reproducibility

Code, tests and raw results are in the author's repository (`collatz/pinball/`, `scripts/`). The computations, the literature search and the drafting of this note were carried out with substantial assistance from Claude (Anthropic); all statements were checked by computation as described.

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